

Ashish S. Joshi

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SUMMARY

Multimodal AI researcher and M.S. Artificial Intelligence candidate focused on vision-language models, spatial reasoning, visual grounding, interpretability, and robustness. Experience designing VLM benchmarks, evaluating multimodal models, analyzing hallucination and failure modes, and developing training-free mitigation methods. Hands-on with Python, PyTorch, Hugging Face, CLIP, Vision Transformers, Qwen2.5-VL, InternVL, LLaVA, and GPU-based model evaluation.

EDUCATION

Boston University

Aug. 2025 - May 2027

Master of Science (MS) in Artificial Intelligence,

Boston, MA

Relevant Courses: Machine Learning, Artificial Intelligence, Multimodal AI, Deep Learning, Natural Language Processing

MIT World Peace University

Jul. 2019 - May 2022

Bachelor of Science (B.Sc.) in Computer Science

Pune, India

Relevant Courses: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming

TECHNICAL SKILLS

Programming & Research – Python, SQL, C++, Git/GitHub, Linux, Jupyter, REST APIs, FastAPI, GPU experimentation

Multimodal AI & VLMs – Vision-Language Models, Qwen2.5-VL, InternVL, LLaVA, BLIP-2, InstructBLIP, CLIP, multimodal reasoning, visual grounding, spatial reasoning

Deep Learning & Vision – PyTorch, TensorFlow, Hugging Face, Vision Transformers, CNNs, embeddings, transfer learning, fine-tuning, OpenCV, Grad-CAM

Interpretability & Evaluation – hallucination analysis, failure analysis, robustness, counterfactual evaluation, ablation studies, groundedness, error analysis, benchmarking

Systems & MLOps – pandas, NumPy, PostgreSQL, Docker, MLflow, CI/CD, experiment tracking, model serving, AWS/GCP

WORK EXPERIENCE

High Energy Materials Research Laboratory (HEMRL), DRDO

Pune, India

Apprentice AI Engineer | AI Project Lead

Feb. 2023 – May 2025

- Led technical execution across **15+ applied AI initiatives**, owning problem formulation, experimental design, model development, evaluation, and technical handoff while coordinating a **3-engineer team** and collaborating directly with scientists.
- Built reusable **computer vision and multimodal experimentation pipelines** for dataset preparation, model training, inference, representation analysis, benchmarking, and experiment tracking using Python, PyTorch, and GPU-based workflows.
- Developed and evaluated **deep learning, transformer, and multimodal AI systems** across visual recognition and reasoning tasks, combining representation learning, transfer learning, systematic ablations, and model comparison.
- Optimized AI systems on **180K+ samples**, improving validation performance by **14%** and reducing inference latency by **28%**; established workflows for robustness, interpretability, failure analysis, hallucination, grounding, and model reliability.
- Productionized research workflows with **Docker, MLflow, CI/CD, FastAPI, and GPU inference**, enabling reproducible experiments, model versioning, deployment, and reliable internal handoff.

StackLab.in

Pune, India

AI/ML Engineer Intern

Apr. 2022 – Dec. 2022

- Built end-to-end **machine learning and computer vision workflows** spanning data ingestion, preprocessing, feature engineering, model development, evaluation, and client-facing delivery using Python, PyTorch, scikit-learn, and OpenCV.
- Integrated trained models into production-facing systems through **REST APIs, SQL-backed services, and external application components**, while performing systematic debugging, error analysis, and performance optimization.
- Owned rapid **build-test-evaluate cycles** in a startup environment, translating changing requirements into deployable ML features through experimentation, model benchmarking, API integration, and iterative development.

PROJECTS

SpatialBench-VLM: Benchmarking Spatial Reasoning in Vision-Language Models

Feb. 2026 – May 2026

- Built a unified VLM evaluation pipeline across **CV-Bench-2D**, **VSR**, **Spatial-MM**, **SpatialMQA**, **SAT**, and **MME-RealWorld-Lite** to study spatial reasoning, object relations, directional understanding, and visual grounding.
- Benchmarked **BLIP-2**, **InstructBLIP**, **Phi-3.5-Vision**, **Cambrian-8B**, and **Qwen2.5-VL**, performing cross-model failure analysis to identify benchmark-specific weaknesses and recurring spatial reasoning errors.

SARF: Structure-Aware Representation Fusion for CLIP

Mar. 2026 – Apr. 2026

- Developed a **structure-aware representation framework** for technical circuit diagrams using frozen CLIP ViT-B/32 encoders with complementary RGB and edge-derived visual representations.
- Evaluated **sum**, **concatenation**, and **learned fusion strategies** to analyze how structural information affects representation quality, classification behavior, and robustness in visually sparse technical imagery.

RARC: Reasoning-Aware Retrieval Controller

Jan. 2026 – Mar. 2026

- Designed an adaptive retrieval controller and executed **768 experiments across 36 configurations**, varying retrieval behavior, context strategy, model configuration, and quantization.
- Evaluated **Llama 3.3 70B** and **Qwen2.5 32B** across groundedness, hallucination, relevance, latency, GPU utilization, and memory to characterize retrieval accuracy–efficiency trade-offs.

CoST-H: Counterfactual Spatial Tracking and Hallucination in VLMs

Jun. 2026 – Jul. 2026

- Designed a **2,800-example benchmark across 7 intervention families** to evaluate spatial-state preservation and hallucination in VLMs under movement, cancellation, swaps, contradictions, and hypothetical interventions.
- Developed **CoST-Gate**, a training-free visual-state reconciliation mechanism combining contextual and visually grounded predictions, improving performance by up to **18.6 percentage points**.

SlideMIL-ViT: Weakly Supervised Learning for Whole-Slide Images

Mar. 2026

- Developed a weakly supervised **multiple-instance learning pipeline** for gigapixel pathology slides using UNI2-h representations and hierarchical slide-to-patient feature aggregation.
- Designed patient-level prediction and evaluation workflows for whole-slide imaging, contributing to a **Top-3 placement at the Boston University MedAI Hackathon**.

PUBLICATIONS

Unleashing the Power of Deep Learning: A Novel Approach for Defect Detection in SRM

Feb. 2026

Ashish Joshi, Kaushik Joshi, Rupali Kute, Sunil Jaiswal, Prerna Mishra
FirePhysChem, Elsevier | DOI: 10.1016/j.fpc.2025.06.005

Designing Retrieval Systems for LLMs: Chunk-Based vs. File-Centric Retrieval

May 2026

S. Kalyanaraman, R. Gupta, **A. S. Joshi**, K. S. Joshi, J. Suryawanshi, V. Jadhav
2026 IEEE (SIEDS) | DOI: 10.1109/SIEDS69358.2026.11540259

CoST-H: Counterfactual Spatial Tracking and Hallucination in VLMs

Under Review

Ashish Joshi, Kaushik Joshi

AAAI Conference on Artificial Intelligence (AAAI 2027)

RiskRoll-RAG: Inference-Time Rollback for Safer Long-Form Retrieval-Augmented Generation

Under Review

Kaushik Joshi, Ashish Joshi

AAAI Conference on Artificial Intelligence (AAAI 2027)

LEADERSHIP & TECHNICAL OUTREACH

- Secured **Top-3 placement** at BU MedAI Hackathon for a weakly supervised vision pipeline using whole-slide image embeddings, hierarchical aggregation, and patient-level evaluation.
- Delivered an invited Science Day lecture to **DRDO scientists** on **Vision-Language Models, multimodal reasoning, interpretability, grounding, hallucination, and AI reliability**, connecting emerging VLM research with practical model failure analysis.
- Presented technical topics spanning **multimodal foundation models, representation learning, model interpretability, agentic AI, retrieval-augmented generation, and trustworthy AI** to scientists and engineers.